

The Single-Person Decision Theory

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This talk

The Single-Person Decision Theory

- decisions under certainty
- decisions under uncertainty

Errors and Biases in Decisions

- errors in preferences
- errors in judgment
- rule-rationality/heuristics

Decisions under certainty

Elements of a decision problem

Actions

- the alternatives from which the decision-maker chooses

Outcomes

- consequences of the actions

Preferences

- the decision-maker's comparisons of the outcomes

Decision under certainty

Outcomes

- a *finite* set of outcomes X

Preferences over outcomes

- $\succsim$ is a weak preference relation of a decision maker on X
- $x \succsim y$: for $x \in X$ and $y \in X$, 'x is at least as good as y'
- $\succsim$ is a binary relation¹ that induces:
 - strict preference relation $\succ$

$$x \succ y \Leftrightarrow x \succsim y \text{ but not } y \succsim x$$

- indifference preference relation $\sim$

$$x \sim y \Leftrightarrow x \succsim y \text{ and } y \succsim x$$

¹A binary relation on a set X is a collection of ordered pairs of elements of X .

Preferences: Consistency/Rationality

$\succsim$ is **consistent/rational** if it possesses the following two properties:

1. **completeness**: for all $x \in X$ and $y \in X$

$$x \succsim y \text{ or } y \succsim x$$

2. **transitivity**: for all $x \in X$, $y \in X$ and $z \in X$

$$x \succsim y \text{ and } y \succsim z \Rightarrow x \succsim z$$

Consistency implies existence of a function $u : X \rightarrow \mathbb{R}$ such that

$$u(x) \geq u(y) \Leftrightarrow x \succsim y.$$

u is a payoff function *over outcomes*.

Let $f : A \rightarrow X$ be mapping from actions to outcomes. Then, $v : A \rightarrow \mathbb{R}$, where $v(a) \equiv u(f(a))$, is a payoff function *over actions*.

Preferences: Consistency/Rationality

- consistent/rational preferences are not necessary selfish
- consistent/rational preferences may involve also 'preferences over actions', i.e., preference over the ways we reach an outcome (to that end, we need to define the 'outcomes' such that it includes also the way it has been achieved)
- consequently, consistent/rational preferences may involve morals and norms

Homo oeconomicus under certainty (Rational paradigm)

1. homo oeconomicus knows her decision problem (A, X, f)
 - knowledge of (i) all possible actions, A , (ii) all possible outcomes, X , (iii) exactly how each action affects which outcome will materialize, f
2. homo oeconomicus has consistent preferences over X ($\succsim$)
 - complete and transitive
3. homo oeconomicus maximizes her payoff (u, v)
 - A player facing a decision problem with a payoff function over actions, v , is *payoff-maximizing* if she chooses an action $a \in A$ that maximizes his payoff. That is, $a^* \in A$ is chosen only if $v(a^*) \geq v(a)$ for all $a \in A$.

Decisions under uncertainty

Decision under uncertainty

Uncertain outcomes

- index certain outcomes in X by $i = 1, \dots, N$
- an uncertain outcome = a (simple) lottery over certain outcomes

Simple lotteries

- $p_i \in [0, 1]$ is an objective (and known) probability of outcome i
- a simple lottery L is a list $L = (p_1, \dots, p_N)$ such that $p_i \geq 0$ for all i and $\sum_i p_i = 1$

= a (discrete) probability distribution

- the set of all simple lotteries is $\mathcal{L}$ (simplex with $N - 1$ dimensions)
- any compound lottery can be reduced into a simple lottery
- ! finite X but infinite $\mathcal{L}$

Decision under uncertainty

Preferences over uncertain outcomes are preferences over simple lotteries.

- $\succsim$ is a weak preference relation on $\mathcal{L}$ (instead of X)
- consistency/rationality (completeness, transitivity)
- + **continuity**: For all $L' \in \mathcal{L}$, the sets $\{L : L \succsim L'\}$ and $\{L : L' \succsim L\}$ are closed sets.

By continuity, *lexicographic* preferences are not allowed.

- Lexicographic preferences mean that individuals disregard 'lower-order' differences.

Decision under uncertainty

Example (Parents)

- strict preferences: $1 \text{ EUR} \succ 0 \text{ EUR} \succ \text{'loss of a child'}$
- equivalently: $(1 - \alpha, 0, \alpha) \succ (0, 1, 0)$ for $\alpha = 0$
- lexicographic (discontinuous) preferences:
 - $(0, 1, 0) \succ (1 - \alpha, 0, \alpha)$ for any $\alpha > 0$
- continuous preferences:
 - $\exists \alpha > 0$ such that $(1 - \alpha, 0, \alpha) \succ (0, 1, 0)$
- Many parents claim to have lexicographic preferences, but in fact don't maximize survival of their kids at 'any cost' (e.g., they travel by car with kids).

Decision under uncertainty

Consistency and continuity of preferences over lotteries implies existence of a (payoff/utility) function $U : \mathcal{L} \rightarrow \mathbb{R}$ such that

$$L \succsim L' \Leftrightarrow U(L) \geq U(L').$$

To compare:

- For the existence of a payoff function under certainty (over a *finite* set of outcomes X), we impose only *consistency*.
- For the existence of a payoff function under uncertainty (over an *infinite* set of lotteries $\mathcal{L}$), we impose *consistency* and *continuity*.

Decision under uncertainty

- **independence:** For any $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0, 1)$:

$$L \succsim L' \Leftrightarrow \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''$$

- intuition: If a third lottery may occur instead of a lottery, its possible occurrence doesn't affect the preference ordering of the two lotteries.
 - ! Lottery L'' occurs *instead* of lottery L or L' . The consumer choosing between L and L' doesn't consume L *with* L'' or L' *with* L'' , but only L'' *instead* of L or L' .
- This puzzled even Paul Samuelson:

I am simply confused... the assumption would impose an arbitrary "straight-jacket" on individual preferences over risky alternatives.... The postulate was a "gratuitously-arbitrary special-implausible hypothesis."
- After discussions with Friedman, Savage and Baumol:

... now I must eat my words. As you know I hate to change my mind, but I hate worse to hold wrong views, and so I have no choice.

Decision under uncertainty

Expected utility theorem (von Neumann-Morgenstern)

Consider a rational and continuous preference relation $\succsim$ on $\mathcal{L}$ that satisfies independence. Then, $\succsim$ is represented by a utility/payoff function U with the expected utility form.

- Utility function $U : \mathcal{L} \rightarrow \mathbb{R}$ has an *expected utility form* if and only if there exists a vector $(u_1, \dots, u_N) \in \mathbb{R}^N$ such that for every $L \in \mathcal{L}$

$$U(L) = p_1 u_1 + \dots + p_N u_N.$$

- u_i is called Bernoulli utility of the outcome i .
- Bernoulli utilities can be changed by any positive ($a > 0$) affine transformation: $\sum_i p_i u_i \geq \sum_i q_i u_i$ is equivalent to $\sum_i p_i (a u_i + b) \geq \sum_i q_i (a u_i + b)$ because $\sum_i p_i = 1 = \sum_i q_i$ and $a > 0$.
- U is a linear function of $L = (p_1, p_2, \dots, p_N)$.

Application: Retrieving payoffs from choices

How to construct u , i.e., a payoff function under certainty?

- We ask homo oeconomicus to compare (x, y, z) ; suppose she prefers $x \succ y \succ z$.
- If we analyze only decisions under certainty, we can impose *any* u such that

$$u(x) > u(y) > u(z).$$

- ! These payoffs do *not* represent decisions under uncertainty.
- If we want to analyze decisions under uncertainty, we need preference over lotteries.

Application: Retrieving payoffs from choices

How to construct U , i.e., a payoff function under uncertainty?

- We ask her to report $\alpha \in (0, 1)$ such that she is indifferent between a lottery $(\alpha, 0, 1 - \alpha)$ over (x, y, z) and a (degenerate) lottery $(0, 1, 0)$:

$$(\alpha, 0, 1 - \alpha) \sim (0, 1, 0)$$

- Using that any positive affine transformation is possible, impose $u_y = 0$.
- Then, (Bernoulli) payoffs must satisfy the constraint:

$$\alpha u_x + (1 - \alpha) u_z = u_y = 0$$

- One solution is $(u_x, u_y, u_z) = (1 - \alpha, 0, -\alpha)$.
- By adding a constant α , another solution is

$$(u_x, u_y, u_z) = (1, \alpha, 0).$$

Ordinal vs. cardinal preferences

How to measure 'intensity' of preferences?

- a long history of discussions about 'measurability' of utility
- to measure 'intensity' of preferences, assume consistency, continuity and independence and find Bernoulli payoffs
- ! 'cardinal' preferences over *certain* outcomes can be constructed if ordinal preferences over *uncertain* outcomes are known

Errors and Biases in Decisions

Errors

Of course, not all of us are cool-blooded like Sherlock Holmes.

- Errors in preferences
- Errors in judgment

[Wikipedia](#): a list of cognitive errors

Errors in preferences over certain outcomes

Suppose your preferences exhibit a cycle:

- $x \succsim y, x \not\succsim y, y \succsim z, z \not\succsim y, z \succsim x, x \not\succsim z$
- a cycle $x \succ y \succ z \succ x$
- intransitive: by transitivity, $x \succsim y$ and $y \succsim z$ imply $x \succsim z$ which is not true

Then, you are vulnerable to a 'money pump' by an arbitrageur (Dutch books).

- Let your preferences over money be separable from preferences over (x, y, z) .
- You will pay $p_{zx} > 0$ for changing x to z .
- You will pay $p_{yz} > 0$ for changing z to y .
- You will pay $p_{xy} > 0$ for changing y to x .
- In total, you will pay $p_{xy} + p_{yz} + p_{zx} > 0$ for nothing.

Questions

Consider 3 outcomes:

- (0, 100 million EUR, 500 million EUR)
- note: lottery probabilities are in percentages
- Question 1: Do you prefer lottery A: (0, 100, 0) or lottery B: (1, 89, 10)?
- Question 2: Do you prefer lottery C: (89, 11, 0) or lottery D: (90, 0, 10)?

The Allais Paradox

- Mas-Collel et al. (1995): the most frequent comparisons are $A \succ B$ and $D \succ C$. This is inconsistent with EUT.
- Expected utility theory (independence, EUT) requires either $A \succ B$ and $C \succ D$, or $B \succ A$ and $D \succ C$.

$$B - A = (+1, -11, +10) = D - C$$

$$U(B) - U(A) = \frac{1}{100} u_0 - \frac{11}{100} u_{100} + \frac{10}{100} u_{500} = U(D) - U(C)$$

- ! Without independence, agents are again potentially vulnerable to ‘money pumping’ (a collection of bets/trades that leaves one party strictly better off and the other party strictly worse off).

The Allais Paradox

How to visualize the preferences over lotteries A, B, C, and D in 2D?

- The set of lotteries over three outcomes is a simplex

$$\Delta^3 \equiv \{p \in \mathbb{R}_+^3 : p_1 + p_2 + p_3 = 1\}.$$

- We can map the simplex Δ^3 into a triangle in 2D space. Consider a space defined by (p_1, p_3) . The simplex maps into a set $\{(p_1, p_3) \in [0, 1]^2 : p_1 + p_3 \leq 1\}$. (p_2 is given residually, $p_2 = 1 - p_1 - p_3$.)
- Utility function in the 2D space is $U_{13}(p_1, p_3) \equiv U(p_1, 1 - p_1 - p_3, p_3)$.
- A level (indifference) curve of U_{13} is a set of pairs $\{(p_1, p_3) \in \mathbb{R}_+^2, p_1 + p_3 \leq 1 : U_{13}(p_1, p_3) = U\}$.

$$p_1 u_1 + (1 - p_1 - p_3) u_2 + p_3 u_3 = U.$$

$$p_3 = \frac{u_2 - u_1}{u_3 - u_2} p_1 + \frac{U - u_2}{u_3 - u_2}$$

- ! The level (indifference) curves are linear in the 2D space.

The Allais Paradox (Blavatsky, Ortmann and Panchenko, 2022)

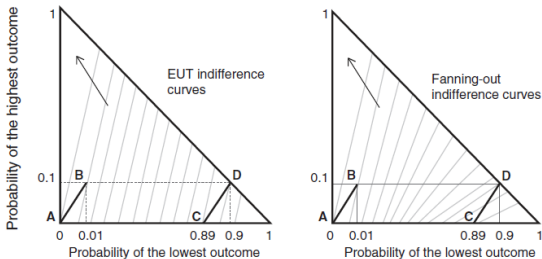


FIGURE 1. ILLUSTRATION OF THE ALLAIS PARADOX IN THE PROBABILITY TRIANGLE

- The Allais Paradox is observed in experiments with high hypothetical payoffs, the medium outcome being close to the highest outcome and when lotteries are presented in reduced (not compound) form.

Errors in preferences over uncertain outcomes

- expected utility theory is often violated
- context matters
- violations often associated with small probabilities and large hypothetical payoffs
- presenting lotteries in compound vs. reduced form also matters

Errors in judgment

Linda problem

- Linda is a student deeply concerned with discrimination and social justice. Is Linda more likely a bank teller or a bank teller and an active feminist?
- But every feminist bank teller is a bank teller.
- *Conjunction fallacy* is when specific conditions are seen as more probable than general conditions.
- Why? Linda problem violates conversational maxims in that people assume that the question obeys the *maxim of relevance*.

Anchoring bias

- Survey respondents observed a roulette wheel that had been fixed to stop at either 65 or 10.
- They were asked to estimate the percentage of UN countries located in Africa.
- The respondents were influenced by irrelevant numbers.
- Anchoring is a cognitive bias where an individual depends too heavily on an initial piece of information offered (anchor) to make subsequent judgments.
- Why? We typically assume that the *initial information provided is relevant*.

Errors in judgment

Asian disease

Imagine that the U.S. is preparing for the outbreak of an unusual Asian disease, which is expected to kill 600 people.

- Program A will save 200 lives.
- Program B will save no-one with probability $\frac{2}{3}$ and 600 lives with probability $\frac{1}{3}$.
- Which program do you prefer?

Now consider another choice.

- Under Program C, 400 people die.
- Under Program D, no-one dies with probability $\frac{1}{3}$ and 600 people die with probability $\frac{2}{3}$.
- Which program do you prefer?

Framing: Most physicians answer $A \succ B$ and $D \succ C$ (Kahneman and Tversky, 1981).

Hot hand fallacy (Miller and Sanjurjo, 2018)

Hot hand belief

- a widespread belief among basketball fans, professional players, coaches and experts that there is *momentum in shooting performance* (“hot hand”)

Gilovich, Vallone, and Tversky (1985) call it a “massive and widespread cognitive illusion” (hot hand fallacy). They illustrate it as follows:

- Take independent Bernoulli trials such as a fair coin.
- Unconditional and conditional win probability are the same ($\frac{1}{2}$).
- They define hot hand fallacy as an (incorrect) belief that conditional probabilities are different from the unconditional probability $\frac{1}{2}$.

Hot hand fallacy

What did Gilovich, Vallone, and Tversky (1985) actually do?

- NBA field goal data, NBA free-throw data, and NCAA collegiate players
 - For each player i , calculate *conditional shooting percentages* $P^i(\text{hit} \mid k \text{ hits})$ (hit the last k shots) and $P^i(\text{hit} \mid k \text{ misses})$ (missed the last k shots).
 - a paired t-test of whether $\mathbb{E}[P^i(\text{hit} \mid k \text{ hits})] - \mathbb{E}[P^i(\text{hit} \mid k \text{ misses})] = 0$
- = a test of whether this stochastic process involves dependent or independent Bernoulli trials
- = a test whether 'recent history matters' by comparing 2 extreme histories
- When the null hypothesis is not rejected, hot hand belief is interpreted as an illusion that the conditional win probability is larger than the unconditional win probability.

Hot hand fallacy: GVT's procedure

Basketball player with 8 shots (a string of length $n = 2$): 1010 0110

- unconditional success probability 50%
- $k = 1$ histories: only the last previous shot matters
- outcomes for the good history (previous shot is 1): 10, 10, 11, 10
- conditional success probability 25%
- outcomes for the bad history (previous shot is 0): 01, 00, 01
- conditional success probability 66%

Hot hand fallacy: A fair coin

We apply the GVT's procedure to a fair coin (where the process is known to be independent Bernoulli trials).

- Let $k = 1$: a 'streak' (of previous hits) is of length 1.
 - Let $n = 2, 3$ the length of a 'string' (a sequence of shots).
 - Define outcomes of interest: The test is interested only in "hot hands" (confirmed wins, HH) and "flips" (not confirmed wins, HT).
 - Instead of testing $\mathbb{E}[P^i(\text{head} \mid 1 \text{ head})] = \frac{1}{2} = \mathbb{E}[P^i(\text{head} \mid 1 \text{ tail})]$, we test whether the GVT's procedure gives $\mathbb{E}[P^i(\text{head} \mid 1 \text{ head})] = \frac{1}{2}$ for a fair coin.
 - For a fair coin, $P^i(\text{head} \mid 1 \text{ head})$ is defined as the proportion of HHs among HHs and HTs in the string i .
- ! This success rate is not defined for strings where both HHs and HTs are absent – we eliminate such strings.

Hot hand fallacy: A fair coin

Example ($n = 2$)

- 4 strings; each with probability $\frac{1}{4}$
- 2 strings where P^i is defined
 - String HH: 1 HH, 0 HT – success rate 100%
 - String HT: 0 HH, 1 HT – success rate 0%
 - String TH: success rate not defined
 - String TT: success rate not defined
- The total number of HHs and HTs is the same (1).
- The expected success rate is 50%.

= equal to the conditional/unconditional win (H) probabilities

Gilovich, Vallone, and Tversky (1985) and subsequent experimental literature implicitly supposes that the expected success rate, $\mathbb{E}[P^i(\text{hit} \mid k \text{ hits})]$, is a measure of the conditional/unconditional win probability of H, $\frac{1}{2}$. But is it correct?

Hot hand fallacy: A fair coin

Example ($n = 3$)

- 8 strings; each with probability $\frac{1}{8}$
- 6 strings where P^i is defined
 - String HHH: 2 HH, 0 HT – success rate 100%
 - String HHT: 1 HH, 1 HT – success rate 50%
 - String HTH: 0 HH, 1 HT – success rate 0%
 - String HTT: 0 HH, 1 HT – success rate 0%
 - String THH: 1 HH, 0 HT – success rate 100%
 - String THT: 0 HH, 1 HT – success rate 0%
 - String TTT: success rate not defined
 - String TTH: success rate not defined
- The total number of HHs and HTs is the same (4).
- But the expected success rate is $\frac{250}{600} = \frac{5}{12} < \frac{6}{12}$.
- ! **not** equal to the conditional/unconditional win (H) probabilities

Hot hand fallacy

Takeaways?

- For $n = 3$, HH appears twice in HHH, but HT never appears twice in any string!
- The expected success rate **underestimates** the win probability.
- The GVT's test is a *biased selection procedure*.

More generally, in strings of finite length:

- The streaks of “hot hands” are **longer** than the streaks of “flips”.
- HH subsequences are more clustered/packed in, because HH **can** overlap with HH, but HT **cannot** overlap with HT.
- HT are more dispersed (across more strings) while HH is more concentrated (across less strings).

What are the implications?

- If a ‘hot hand’ is understood as a higher relative **concentration** of HHs within a string (instead of a higher conditional hit probability), then “hot hand” belief is a correct belief.
- Using NBA data, Miller and Sanjurjo (2018) find that upon correcting for the selection bias, the conclusions of the GVT’s study disappear.

Rule rationality

Rule rationality (Aumann, 2019)

Key ideas:

- Act-rationality: Agents optimize over actions.
- Rule-rationality: Agents over rules/heuristics.

Why heuristics?

- Limited cognitive abilities = a limited number of heuristics

What are implications?

- Heuristics are optimal only 'on average'.
 - ! Optimal heuristics/rules are context-dependent (depends on what is 'typical').
- In unusual situations, heuristics fail (even dramatically).
- But if failures become 'typical situations', the heuristics will adapt.
- Certain industries (e.g., advertising) tend to systematically exploit failures.

Example: Overeating

- heuristic 'eat when you have appetite'
- unusual situation: sedentary nature of modern life

Ambiguity Aversion

- two outcomes: good and bad
- $p \in (0, 1)$ is probability of the good outcome
- *unambiguous* lottery: p is known; for example, a simple lottery with $p = \frac{1}{2}$
- *ambiguous* lottery: p is unknown and we have *no argument* why it should be of specific value

How to perceive an ambiguous lottery?

- a compound lottery of all p -lotteries where each simple p -lottery has *equal* probability
- = principle of insufficient reason
- a reduced lottery of this compound lottery is a simple lottery with $p = \frac{1}{2}$

What is experimentally observed?

- Decision-makers prefer an unambiguous lottery with known $p = \frac{1}{2}$ to the ambiguous lottery which can be reduced to $p = \frac{1}{2}$.
- Why? (i) The principle of insufficient reason is not used or probabilities are miscalculated. (ii) Dislike of 'lack of knowledge' (*ambiguity aversion*) even if it is payoff-irrelevant (e.g., due to cognitive costs of using compound lotteries). Or something else?

Ellsberg experiment

Bags

- White bag (● ○ ○): 1 red ball and 2 white balls
- Black bag (● ● ●): 1 red ball and 2 black balls
- Nature has chosen a white bag or a black bag, with **unknown** probability.
- You don't observe color of the bag.

Bets

- You may bet on a red ball (Red bet) or a white ball (White bet).
- If you win your bet, you get 1. If you lose, you get 0.
- Red bet: an *unambiguous* lottery with $p = \frac{1}{3}$
- White bet: an *ambiguous* lottery reduced to $p = \frac{1}{3}$

Ellsberg experiment

Which bet is better?

- By principle of insufficient reason, we can set the probabilities of bags to $(\frac{1}{2}, \frac{1}{2})$.
 - 50%: White bag (● ○ ○): 1 red ball and 2 white balls
 - 50%: Black bag (● ● ●): 1 red ball and 2 black balls
 - Red and white bets are different compound lotteries over different simple lotteries, but they give an **identical** reduced lottery $L = (\frac{4}{6}, \frac{2}{6})$ over payoffs (0, 1).
- = Simply, each bet is on 2 out of 6 equally likely balls (● ● ○ ○ ● ●).

Ellsberg experiment revisited

Consider two modifications.

- The decision-maker expects **another draw** from the same bag.
- The decision-maker is risk-averse. (This was irrelevant in the original Ellsberg experiment because only two outcomes exist for 1 draw: 0 or 1. Now, we have three outcomes for 2 draws: 0, 1 or 2.)

We will analyze 2 settings:

- double draw **without** a revision: the second bet is identical to the initial bet
- double draw **with** a revision: the second bet can be different from the initial bet

Double draw without a revision of the initial bet

- Red bet: $L_r = (\frac{4}{9}, \frac{4}{9}, \frac{1}{9})$ over $(0, 1, 2)$ outcomes

$$\Pr(0 \text{ red}) = \frac{1}{2} \left(\frac{2}{3} \frac{2}{3} \right) + \frac{1}{2} \left(\frac{2}{3} \frac{2}{3} \right) = \frac{4}{9}$$

$$\Pr(1 \text{ red}) = \frac{1}{2} \left(\frac{2}{3} \frac{1}{3} + \frac{1}{3} \frac{2}{3} \right) + \frac{1}{2} \left(\frac{2}{3} \frac{1}{3} + \frac{1}{3} \frac{2}{3} \right) = \frac{4}{9}$$

$$\Pr(2 \text{ red}) = \frac{1}{2} \left(\frac{1}{3} \frac{1}{3} \right) + \frac{1}{2} \left(\frac{1}{3} \frac{1}{3} \right) = \frac{1}{9}$$

- White bet: $L_w = (\frac{5}{9}, \frac{2}{9}, \frac{2}{9})$ over $(0, 1, 2)$ outcomes

$$\Pr(0 \text{ white}) = \frac{1}{2} 1 + \frac{1}{2} \left(\frac{1}{3} \frac{1}{3} \right) = \frac{5}{9}$$

$$\Pr(1 \text{ white}) = \frac{1}{2} 0 + \frac{1}{2} \left(\frac{2}{3} \frac{1}{3} + \frac{1}{3} \frac{2}{3} \right) = \frac{2}{9}$$

$$\Pr(2 \text{ white}) = \frac{1}{2} 0 + \frac{1}{2} \left(\frac{2}{3} \frac{2}{3} \right) = \frac{2}{9}$$

- The expected number of balls in both lotteries is **identical**:

$$\frac{4}{9} 1 + \frac{1}{9} 2 = \frac{6}{9} = \frac{2}{9} 1 + \frac{2}{9} 2$$

- But White bet is **more risky**.

! The risk-averse expected-utility maximizer prefers the Red bet.

Double draw with a revision of the initial bet

- If the initial bet is Red, revise the bet (strictly) only if you observe the white ball.
 - red ball in Draw 1: beliefs about bags don't change, keep Red bet

$$\Pr(\text{red, red}) = \frac{1}{2} \left(\frac{1}{3} \frac{1}{3} \right) + \frac{1}{2} \left(\frac{1}{3} \frac{1}{3} \right) = \frac{1}{9}$$

$$\Pr(\text{red, white}) = \frac{1}{2} 0 + \frac{1}{2} \left(\frac{1}{3} \frac{2}{3} \right) = \frac{1}{9}$$

$$\Pr(\text{red, black}) = \frac{1}{2} \left(\frac{1}{3} \frac{2}{3} \right) + \frac{1}{2} 0 = \frac{1}{9}$$

- black ball in Draw 1: bag is black, keep Red bet

$$\Pr(\text{black, red}) = \frac{1}{2} \left(\frac{2}{3} \frac{1}{3} \right) + \frac{1}{2} 0 = \frac{1}{9}$$

- white ball in Draw 1: bag is white, choose **White bet**

$$\Pr(\text{white, white}) = \frac{1}{2} 0 + \frac{1}{2} \left(\frac{2}{3} \frac{2}{3} \right) = \frac{2}{9}$$

- Initial Red bet: $L'_r = \left(\frac{3}{9}, \frac{5}{9}, \frac{1}{9} \right)$ over $(0, 1, 2)$

$$\Pr(1 \text{ ball}) = 3 \frac{1}{9} + \frac{2}{9} = \frac{5}{9}$$

$$\Pr(2 \text{ balls}) = \frac{1}{9}$$

- The expected number of balls is $\frac{7}{9}$.

Double draw with a revision of the initial bet

- If the initial bet is White, revise the bet (strictly) only if you observe the black ball.

- red ball in Draw 1: beliefs about bags don't change, keep White bet

$$\Pr(\text{red, white}) = \frac{1}{2}0 + \frac{1}{2}\left(\frac{1}{3}\frac{2}{3}\right) = \frac{1}{9}$$

- white ball in Draw 1: bag is white, keep White bet

$$\Pr(\text{white, red}) = \frac{1}{2}0 + \frac{1}{2}\left(\frac{2}{3}\frac{1}{3}\right) = \frac{1}{9}$$

$$\Pr(\text{white, white}) = \frac{1}{2}0 + \frac{1}{2}\left(\frac{2}{3}\frac{2}{3}\right) = \frac{2}{9}$$

- black ball in Draw 1: bag is black, choose Red bet

$$\Pr(\text{black, red}) = \frac{1}{2}\left(\frac{2}{3}\frac{1}{3}\right) + \frac{1}{2}0 = \frac{1}{9}$$

- Initial White bet: $L'_w = (\frac{4}{9}, \frac{3}{9}, \frac{2}{9})$ over $(0, 1, 2)$

$$\Pr(1 \text{ ball}) = 3\frac{1}{9} = \frac{3}{9}$$

$$\Pr(2 \text{ balls}) = \frac{2}{9}$$

- The expected number of balls is **identical**, $\frac{7}{9}$. But White bet is **more risky**.
! The risk-averse expected-utility maximizer prefers the Red bet.

Ellsberg experiment revisited

- In our example, the decision-maker expects the second draw with probability 1.
- The result extends to any probability $\epsilon \in (0, 1)$.
- Any ϵ -uncertainty over the future *unrevised* draw coupled with risk aversion implies that the Red bet is optimal.
- Any ϵ -uncertainty over the future *revised* draw coupled with risk aversion implies that the Red bet is optimal.
- The decision to choose a Red bet in Ellsberg experiment is not necessarily due to ambiguity aversion, but due to (i) the ϵ -uncertainty over the future draw, or (ii) the use of an optimal decision heuristic (rule) that accounts for the possibility of a dynamic (instead of static) decision environment where the state of the world (i.e., chosen bag) is persistent.

Key lessons

- Homo oeconomicus is a hypothetical decision-maker with consistent, continuous and independent preferences that chooses her most preferred outcome in her (known) decision problem.
 - Consistent, continuous and independent preferences over lotteries are represented by an expected utility function over Bernoulli utilities. Bernoulli utilities are unique up to a positive affine transformation.
 - Equivalently, we say homo oeconomicus is an *expected-utility maximizer*.
 - Homo oeconomicus can be altruistic.
 - Homo oeconomicus can be moral (norms constitute part of outcomes, and preferences are over both consumption and norms).
 - Homo sapiens sapiens is often a different animal:
 - inconsistency because of incomplete preferences
 - discontinuity because of lexicographic preferences
 - preferences violate independence
 - imperfect maximization because of cognitive biases and imperfect heuristics
- ! When 'irrationality'/error/bias is observed, it is often hard to identify which specific assumption behind homo oeconomicus is violated.

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